

Ancient life

All fossils preserve evidence of ancient life. They can range from tiny microbes to skeletons of giant dinosaurs. Some are spectacular, while others can only be identified using special equipment. But it is often the smallest fossils that tell us the most, helping us date rocks and identify the positions of ancient continents and oceans.

Trilobite

Armored crawler

Type of organism Marine arthropod

Geological period Cambrian to Permian

Age range 520–252 million years old

Location Worldwide

▼ MASS GRAVE

These fossilized trilobites probably died when their environment dried out.

Resembling modern-day horseshoe crabs, these armored sea creatures were some of the earliest complex life forms to be preserved as fossils. They lived on seabeds, where they probably used their insectlike compound eyes to locate and catch other animals for food.



Diplomystus

Freshwater herring

Type of organism Fish

Geological period Eocene

Age range 56–34 million years old

Location North America, South America, Middle East, Africa

This widespread freshwater fish lived in the era that followed the extinction of the giant dinosaurs. It is a distant relative of herrings and sardines, with an upward-facing mouth. It ate small fish and insects living on or near the water surface.



Archaeocidaritis

Spiny sea urchin

Type of organism Echinoid

Geological period Late Devonian to Late Permian

Age range 395–300 million years old

Location North Africa, Europe, North America, Russia

This creature had long, sharp spines on its shell. It moved around on flexible “tube feet” like those of starfish. As with many fossils, this specimen has been crushed by the weight of the rock layers above it.

